

Final Report : Healthcare marketplace

1. Project Overview and Goals

American healthcare is a complex and often frustrating system. The goal of this project is to look at American healthcare and build a dashboard that shows multiple insights into the American healthcare system, and allows for users to search for a plan, and get estimates on how much that would cost.

2 Dataset Description

The current dataset is this [kaggle dataset on health insurance](#). This dataset has a total of 8 tables, with each table being ~1.5gb. In total the dataset is ~12gb of data.

- Characteristics: The data is not clean, and features many rows which are missing multiple values.
- There are also eight total different tables, meaning this adds complexity in the form of multiple joins. Most of the tables have foreign keys.
- The data is structured.

This data is very unclean, and contains several errors, and issues.

3 Description of pipeline workflow and services used

This pipeline workflow used mainly BigQuery and Lookerstudio. This is because these tools are part of the google cloud infrastructure, and are able to handle large data relatively quickly.

BigQuery:

- Data Ingestion
- Data Cleaning
- Data Warehousing

Lookerstudio:

- Data visualization

Data is ingested via the bq load command from .csv files, and once it has been loaded, the data is then cleaned via several BigQuery scripts.

4 Data Processing Steps

There are several steps, these are all required to ensure that the data pipeline is clean, and working properly. **For more detailed instructions you can read the readme.md.** Available from GitHub.

1. **Download and extract .csv files.** This is done via [the following link on kaggle](#), these should be extracted to a folder. If you are using .git, these files should be added to the .gitignore.
2. **Create a Google Cloud Project.** This can be done from the console or from the google cloud gui. Write down the project id, as it will be useful later.
3. **Data Ingestion.** This is done through a shell script, which runs multiple BigQuery commands in series.
4. **Running the CleanBenifitsCostSharing.sql.** This is a series of bigquery commands which clean the CleanBenifitsCostSharing table.
 - a. Converts coinsurance to a double.
 - b. Converts is_covered to booleans.
 - c. Feature extraction to determine the benefit name is adult or child.
5. **Resolving the MillionDollarRates.** This is a series of bigquery scripts to remove the million dollar rates error.
6. **Running the BigQuery Analytical Queries Script.** This is a script which runs multiple analytical queries on the dataset. This is all of the queries in the lookerstudio.

5 Results Obtained

Multiple results were obtained, these are stored in the analytical queries script. These are detailed below in order.

```
SELECT *
FROM `final-project-492620.final_project.Rate` AS r
JOIN `final-project-492620.final_project.Network` AS n
ON r.IssuerId = n.IssuerId
LIMIT 1000;
```

Row	BusinessYear	StateCode	IssuerId	SourceName
1	2014	NE	50022	SERFF
2	2014	NE	50022	SERFF
3	2014	NE	50022	SERFF
4	2014	NE	50022	SERFF
5	2014	NE	50022	SERFF
6	2014	NE	50022	SERFF
7	2014	KS	81711	SERFF

Query 1: This joins the rates and networks tables, these are used for the table where users search for the plans and compare providers.

```

SELECT NetworkName ,
COUNT(*) AS num_plans ,
AVG(IndividualRate) AS avg_rate ,
FROM `final-project-492620.final_project.Rate` AS r
JOIN `final-project-492620.final_project.Network` AS n
ON r.IssuerId = n.IssuerId
GROUP BY NetworkName
ORDER BY num_plans DESC
LIMIT 1000;

```

Job information Results Visualization JSON Execution				
Row	NetworkName	num_plans	avg_rate	
1	Delta Dental	2761816	27.11456681763...	
2	Blue Choice PPO	2442922	359.3771515832...	
3	Business Advantage Network	2082144	493.3682285375...	
4	Renaissance Dental	1834496	26.91396448942...	
5	BlueCare Dental PPO	1817184	357.7923403738...	
6	PPO	1409486	445.2148331012...	

Query 2: Gathers the top 1,000 providers, and then it sorts them by the number of plans they have. It also takes the average healthcare plan cost.

```

SELECT IndividualRate , StateCode
FROM `final-project-492620.final_project.Rate`
ORDER BY IndividualRate DESC

```

Query 3: Ordering the states by the state, This is used for the map of the united states in one of the dashboards.

```

SELECT
BenefitName ,
COUNT(*) AS num_plans ,
AVG(IndividualRate) AS avg_rate ,
FROM `final-project-492620.final_project.Rate` AS r
JOIN `final-project-492620.final_project.CleanBenefitsCostSharing` AS cbc
ON cbc.PlanId
= r.PlanId
GROUP BY BenefitName

```

```
ORDER BY num_plans DESC , avg_rate ASC
LIMIT 2000;
```

Row	BenefitName	num_plans	avg_rate
1	Orthodontia - Adult	58375855	375.6691033562...
2	Dental Check-Up for Children	58361687	375.7543323594...
3	Accidental Dental	58361687	375.7543323594...
4	Routine Dental Services (Adult)	58361687	375.7543323594...
5	Major Dental Care - Adult	58361687	375.7543323594...
6	Basic Dental Care - Adult	58361687	375.7543323594...
7	Basic Dental Care - Child	58361687	375.7543323594...

Query 4: Find the number of plans for each benefit name, and the average rates for those plans.

```
SELECT
BenefitName ,
NetworkName ,
COUNT(*) AS num_plans ,
AVG(IndividualRate) AS avg_rate ,
MIN(IndividualRate) AS min_rate ,
MAX(IndividualRate) AS max_rate
FROM `final-project-492620.final_project.Rate` AS r
JOIN `final-project-492620.final_project.CleanBenefitsCostSharing` AS cbc
ON cbc.PlanId = r.PlanId
JOIN `final-project-492620.final_project.Network` AS net
ON r.IssuerId = net.IssuerId
GROUP BY BenefitName , NetworkName;
```

Row	BenefitName	NetworkName	num_plans	avg_rate	min_rate
1	Routine Dental Services (Adult)	Dentegra Dental PPO	4309786	32.73589193987...	9.53
2	Basic Dental Care - Adult	DentegraStar	194212	35.21456310629...	17.77
3	Basic Dental Care - Child	DentegraStar	194212	35.21456310629...	17.77
4	Major Dental Care - Adult	NetworkBlue	7567230	495.0280543329...	96.63
5	Hearing Aids	NetworkBlue	7567230	495.0280543329...	96.63
6	Hearing Aids	BlueSelect	8105062	500.9386372824...	96.63

Query 5: For each benefit name and each network provider, find the avg, min, and max. This is used in the plan searcher for the dashboard.

```

SELECT StateCode,
AVG(IndividualRate) AS individual_rate ,
AVG(PrimarySubscriberAndOneDependent) AS
Avg_PrimarySubscriberAndOneDependent,
    AVG(PrimarySubscriberAndTwoDependents) AS
Avg_PrimarySubscriberAndTwoDependents,
    AVG(PrimarySubscriberAndThreeOrMoreDependents) AS
Avg_PrimarySubscriberAndThreeOrMoreDependents,
    AVG(CoupleAndOneDependent) AS Avg_CoupleAndOneDependent,
    AVG(CoupleAndTwoDependents) AS Avg_CoupleAndTwoDependents,
    AVG(CoupleAndThreeOrMoreDependents) AS
Avg_CoupleAndThreeOrMoreDependents
FROM `final-project-492620.final_project.Rate`
GROUP BY StateCode
LIMIT 50;

```

StateCode	individual_rate	Avg_PrimarySubs...	Avg_PrimarySubs...	Avg_PrimarySubs...	Avg_CoupleAndO...
WY	403.2936649559...	51.014	73.53188888888...	100.4293333333...	73.46655555555...
NE	258.3728071138...	46.4663257575...	69.0612878787...	84.8676136363...	69.5567424242...
WI	477.0376881961...	60.17299999999...	89.16688125	108.8208687499...	89.26395625
LA	350.6191674426...	49.45251179245...	73.34012971698...	93.81701650943...	74.32198113207...
AK	631.7880306896...	87.61494623655...	107.2298924731...	128.6720967741...	122.6688709677...
OR	277.1508622554...	null	null	null	null
ID	318.5359883840...	52.70821428571...	79.31428571428...	119.2207142857...	78.86571428571...

Query 6: Get the average rate for each customer grouping for each state.

```

SELECT Age ,
AVG(IndividualRate) AS individual_rate ,
AVG(PrimarySubscriberAndOneDependent) AS
Avg_PrimarySubscriberAndOneDependent,
    AVG(PrimarySubscriberAndTwoDependents) AS
Avg_PrimarySubscriberAndTwoDependents,
    AVG(PrimarySubscriberAndThreeOrMoreDependents) AS
Avg_PrimarySubscriberAndThreeOrMoreDependents,
    AVG(CoupleAndOneDependent) AS Avg_CoupleAndOneDependent,
    AVG(CoupleAndTwoDependents) AS Avg_CoupleAndTwoDependents,
    AVG(CoupleAndThreeOrMoreDependents) AS
Avg_CoupleAndThreeOrMoreDependents
FROM `final-project-492620.final_project.Rate`

```

```
GROUP BY Age
ORDER BY Age ASC
LIMIT 1000;
```

Row	Age	individual_rate	Avg_PrimarySubs...	Avg_PrimarySubs...	Avg_PrimarySubs...	Avg_CoupleAndO.
1	0-20	122.2559431841...	null	null	null	nu
2	21	222.8256700133...	null	null	null	nu
3	22	222.8925171511...	null	null	null	nu
4	23	222.9718206531...	null	null	null	nu
5	24	223.0653395413...	null	null	null	nu
6	25	223.6134313993...	null	null	null	nu

Query 7: For each age, this gets the average monthly rate for that age.

```
-- Looking at coins over the years
SELECT CoinsInnTier1 , BusinessYear
FROM `final-project-492620.final_project.CleanBenefitsCostSharing`;
```

Row	CoinsInnTier1	BusinessYear
5048401	0.8	2016
5048402	0.85	2016
5048403	1.0	2016
5048404	1.0	2016
5048405	1.0	2016

Query 8: Looks are the coinsurance over the years.

```
SELECT BusinessYear ,
MAX(CoinsInnTier1) AS max_coins,
APPROX_QUANTILES(CoinsInnTier1, 4)[OFFSET(1)] AS first_quartile_coins ,
AVG(CoinsInnTier1) AS avg_coins,
APPROX_QUANTILES(CoinsInnTier1, 4)[OFFSET(3)] AS third_quartile_coins ,
MIN(CoinsInnTier1) AS min_coins
FROM `final-project-492620.final_project.CleanBenefitsCostSharing`
GROUP BY BusinessYear;
```

Row	BusinessYear	max_coins	first_quartile_coins	avg_coins	third_quartile_coi...	min_coins
1	2015	1.0	0.0	0.116635076855...	0.2	0.0
2	2016	1.0	0.0	0.116662100116...	0.2	0.0
3	2014	1.1	0.0	0.124577716310...	0.2	0.0

Query 9: for each year, this creates a “box-and-whiskers” plot for the coinsurance.

```
SELECT
word,
COUNT(*) AS frequency
FROM
`final-project-492620.final_project.CleanBenefitsCostSharing`,
UNNEST(SPLIT(LOWER(BenefitName), ' ')) AS word
WHERE
word != '' AND word != 'services' AND word != 'care' AND word != 'and' AND
word != 'for' AND word != '-'
GROUP BY
word
ORDER BY
frequency DESC;
```

Row	word	frequency
1	dental	587330
2	outpatient	404092
3	drugs	342227
4	inpatient	337947
5	surgery	335159
6	routine	274652

Query 10: Looking to see what the frequency of each word in the benefit name is.

```
SELECT
BenefitName,
COUNT(*) AS frequency
FROM
```

```
`final-project-492620.final_project.CleanBenefitsCostSharing`
```

```
GROUP BY
```

```
BenefitName
```

```
ORDER BY
```

```
frequency DESC;
```

BenefitName ▼	frequency ▼
Orthodontia - Adult	77377
Basic Dental Care - Adult	77353
Accidental Dental	77353
Dental Check-Up for Children	77353
Major Dental Care - Adult	77353
Routine Dental Services (Adult)	77353
Basic Dental Care - Child	77353

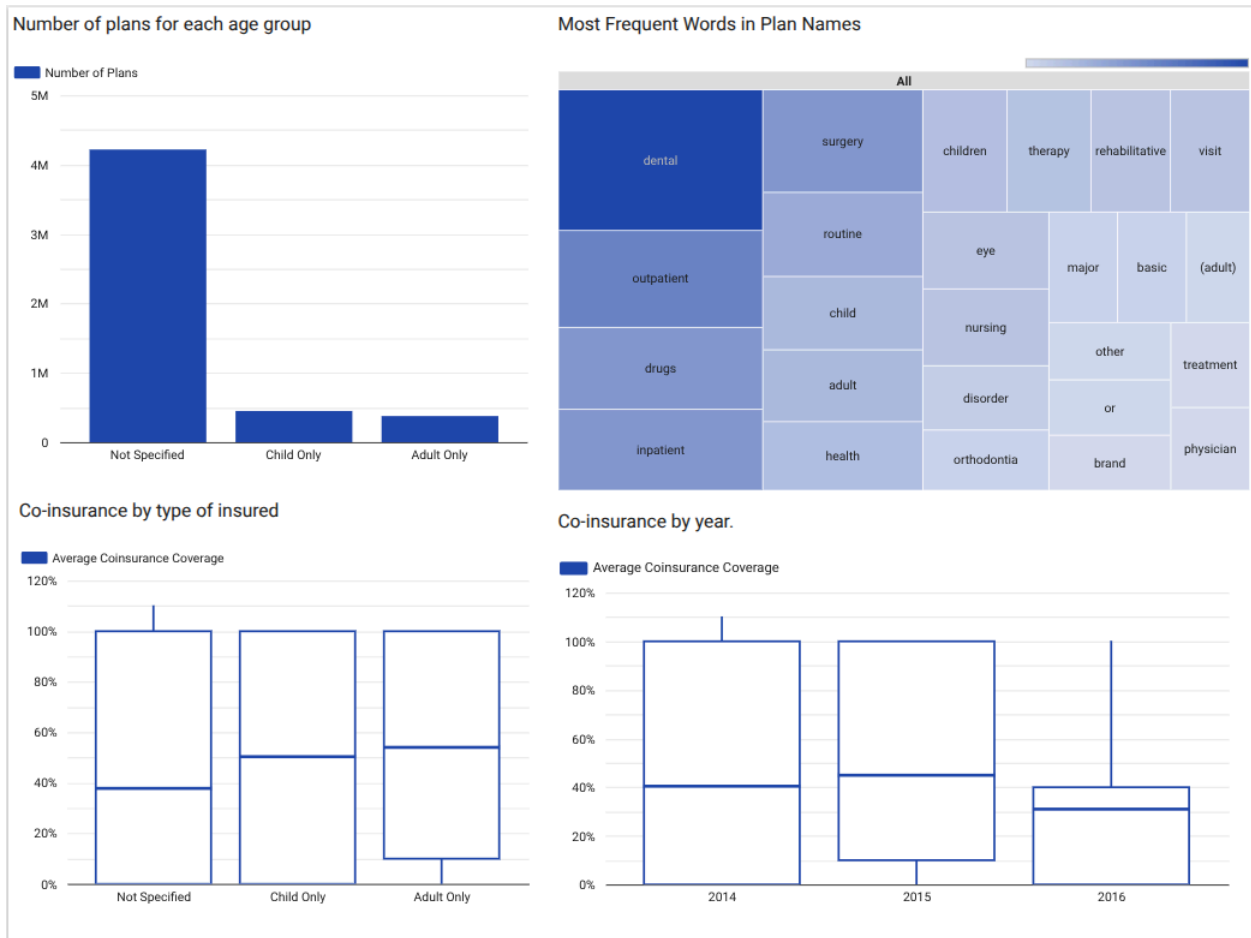
Query 11: For each full benefit name, what is the count of the number of plans for that full benefit name.

```
SELECT AgeGroupType , COUNT(*) AS countOf,
MAX(CoinsInnTier1) AS max_coins,
APPROX_QUANTILES(CoinsInnTier1, 4)[OFFSET(1)] AS first_quartile_coins ,
AVG(CoinsInnTier1) AS avg_coins,
APPROX_QUANTILES(CoinsInnTier1, 4)[OFFSET(3)] AS third_quartile_coins ,
MIN(CoinsInnTier1) AS min_coins
FROM `final-project-492620.final_project.CleanBenefitsCostSharing`
GROUP BY AgeGroupType
LIMIT 100;
```

Row	AgeGroupType ▼	countOf ▼	max_coins ▼	first_quartile_coins ▼	avg_coins ▼
1	Adult Only	377870	1.0	0.0	0.163396883626...
2	Child Only	450807	1.0	0.0	0.145087320948...
3	Not Specified	4219731	1.1	0.0	0.115728210752...

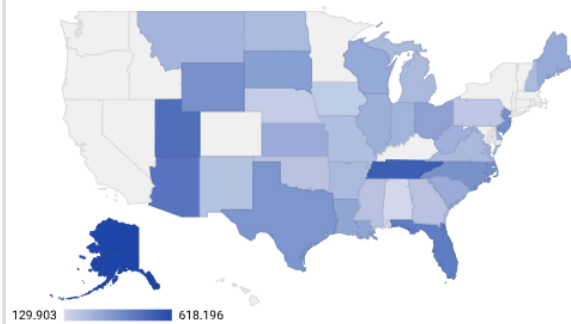
Query 12: Making a “box-and-whiskers” plot for coinsurance by the AgeGroupType, which was a feature I extracted earlier.

6 Visualisations / Dashboard



Dashboards 1: A dashboard which shows the most common words in benefits names, as well as the distribution of Coinsurance for years, and for age groups. It also shows the number of child only, and adult only plans.

Average Individual Rate by each state



Controls and Drill Down

Tobacco -

StateCode -

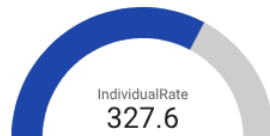
BusinessYear -

Couples:
On average, Plan rates for a single applicant are much higher than for a couple applying.

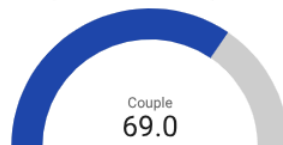
Age:
As expected, individual rates go up the older a person is. This is because health insurance costs go up as you age.

Tobacco:
Smoking kills. Tobacco products decrease your life expectancy, which increases your health care cost.

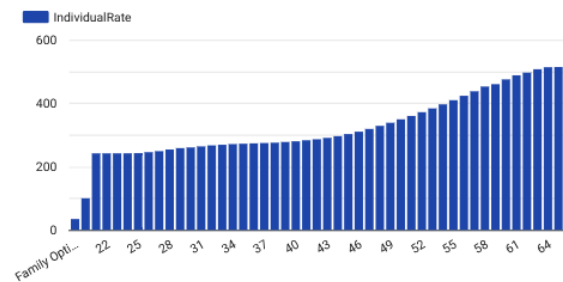
Average Plan Rate for Single



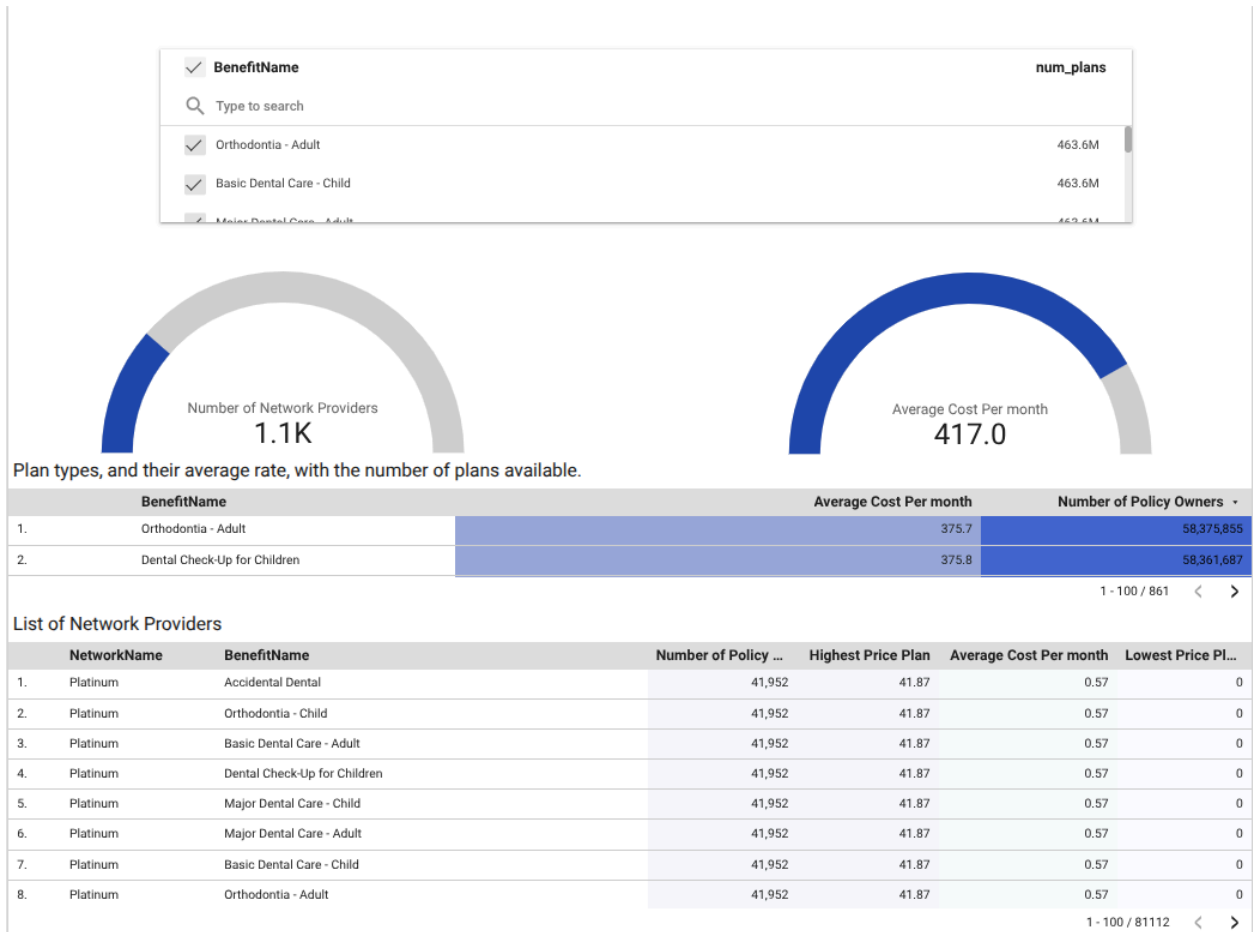
Average Plan rate for couples



Average Plan rates by age.



Dashboard 2: This dashboard shows the average individual rates by state. Note that some states are excluded, because this dataset does not include those states. This also shows the average individual rate per person, and the average rate for couples. (Note: this is per person). This also shows the individual rate trend by age. It also allows the user to filter by smoker, state, or business year.



Dashboard 3: This dashboard allows for users to search by benefit name, and select based on what provider costs the bet on average. They can also see the amount of other policy owners for that benefit name.

I created a website which allows for users to view and switch between these dashboards. This is by making use of iframes, and the GitHub pages website.

[LINK TO THE WEBSITE](#)

Important Health Insurance Terms

Deductible

The fixed amount you pay out-of-pocket before your insurance coverage begins to contribute.

Coinsurance

Your share of costs (e.g., 20%) for a service after you've already met your deductible.

Monthly Rate

The amount of money you pay each month for that specific healthcare plan.

Common Coverage Sections

Dental

Covers routine cleanings and more intensive procedures like fillings or oral surgery.

Vision

Focuses on annual eye exams and discounts or allowances for glasses and contacts.

Mental Health

Provides coverage for therapy, psychiatric evaluations, and substance abuse treatments.

Search for a Plan Overview of Plans by Age and State Coinsurance

Website Screenshot 1: This shows a screenshot of the top half of the website, which shows some very popular healthcare coverage desires, and what they each mean. It also details important healthcare terms on the left.



Website screenshot 2: This shows a screenshot of the website. Note that the tabs at the top of the website allow you to switch between dashboards.

7 Challenges

There were several challenges in this project. The vast majority of these issues were related to data cleaning.

The BenefitsCostSharing table was particularly problematic, this table contains information about the benefit names, and the Coinsurance rates. In particular, the Coins column is a string column, and it needs to be converted to a float. This required a lot of CASE and WHEN statements.

Row	BenefitName	BusinessYear	CoinsInnTier1
4	Skilled Nursing Facility	2015	10% Coinsurance after deductible
5	Skilled Nursing Facility	2015	10% Coinsurance after deductible
6	Skilled Nursing Facility	2015	No Charge after deductible

Fig 1: Screenshot of the CostBenefitsSharing table prior to cleaning.

I also spent some time working on the Rates table, which contains several “one million dollar per month” insurance rates. Figuring out how best to deal with these was difficult. On one hand, this dataset is from the government, which is likely a reputable data source. On the other hand, no human could reasonably expect another to pay one million dollars per month.

There are about 47,790 of these entries, meaning they made up less than 0.3% of the table. These values were ultimately removed, as they seemed like they must be a mistake of some sort.

Job information		Results	Visualiz
w	num_plans	IndividualRate	
1	47790	999999.0	

Fig 2: A screenshot of the number of plans with rates about one million.

8 Lessons Learned

Overall I learned how to use GoogleBigQuery on a long term project, and I learned how to manage and warehouse data on the cloud. I sharpened my SQL skills, and I also learned more about OLAP operations.

In particular, I learned a lot more about SQL CASE queries, which became very useful when I was performing the data cleaning.

9 Potential Next Steps

If I had more time, I'd like to add some dropdowns to the Benefit Names search to add filters for age and tobacco. I'd like age for this to be some sort of slider. This would allow for the user to find the best rates for the plan that they are looking at.

I would also like to add the amount of Coinsurance for the search bar dashboard. I think that would allow for users to better compare plans.

I am also unable to fully process the non-percent copays, which is where copay is listed as only an amount. I then replaced these numbers with the "average percent coverage for a non-percent copay". This was impossible to translate into a percentage, and if I had more time, I'd like to filter this to be its own column, with its own analytics.